Aleatoric Uncertainty Medical Image Segmentation Estimation via Flow Matching

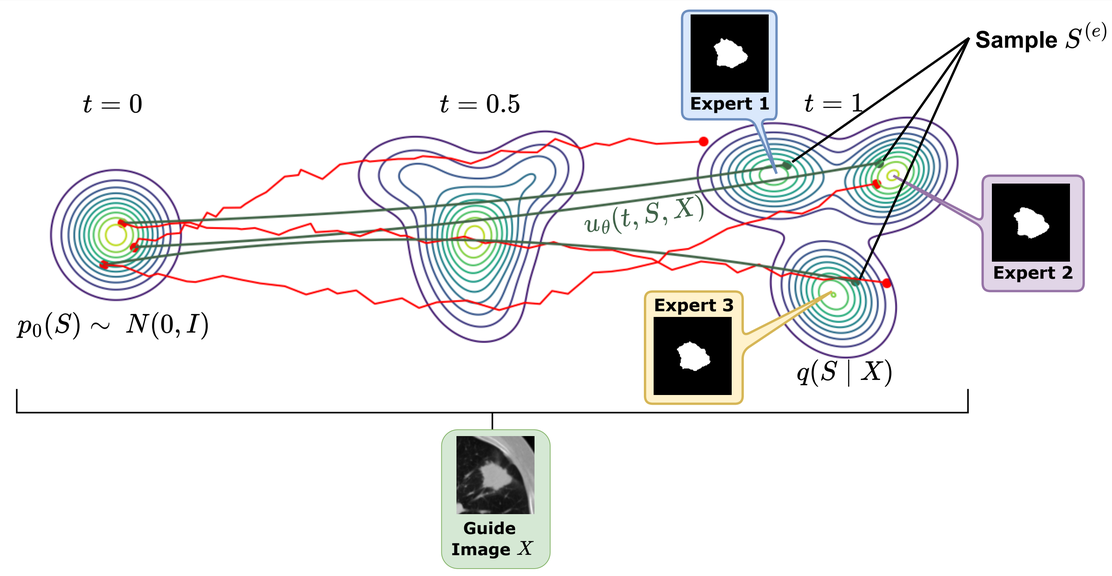
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OVERVIEW

- Quantifying **aleatoric uncertainty** is essential to reflect **inter-annotator variability** in medical image segmentation
- Diffusion-based generative models approximate data distributions but struggle with **stochastic sampling** and inexact density estimation

→ **Conditional flow matching provides accurate segmentations and reliable uncertainty maps by learning exact densities.**



CONTRIBUTIONS

- Introduce a **conditional flow matching** framework that learns a **deterministic velocity field** from input images and expert annotations to estimate **aleatoric uncertainty**.
- Generate segmentation samples that align with anatomical context and capture variability among annotators.

METHOD

- Given a medical segmentation dataset:

$$\mathscr{D} = \left\{ \left(X_i, \left\{ S_i^{(1)}, S_i^{(2)}, ..., S_i^{(E)} \right\} \right) \right\}_{i=1}^{N}$$

- The flow ψ_t that evolves a source sample $S_0 \sim p_0(S)$:

$$\frac{d}{dt}\psi_t(S_0) = u_\theta(t, \psi_t(S_0), X), \quad \psi_0(S_0) = S_0$$

METHOD (cnt.)

- Linear interpolation: $S_t = (1-t)S_0 + t, S^{(e)}$
- Target velocity: $u\left(t, S_t \mid S^{(e)}, X\right) = \dfrac{S^{(e)} - S_0}{1-t}$
- The conditional flow matching (CFM) loss:

$$\mathscr{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, S_0, S^{(e)}, X}\left[\left(u_\theta(t, S_t, X) - \frac{S^{(e)} - S_0}{1-t} \right)^2 \right]$$

- The guided velocity:

$$u_\theta^{\text{guided}}(t, S, X) = u_\theta(t, S, X) + w[u_\theta(t, S, X) - u_\theta(t, S)]$$

- Generate multiple segmentation samples from $\left\{ S_0^{(i)} \right\}_{i=1}^{M}$ and Variability is measured by pixel-wise variance: $S_1^{(i)} = \psi_1\left(S_0^{(i)}; X\right)$

RESULTS

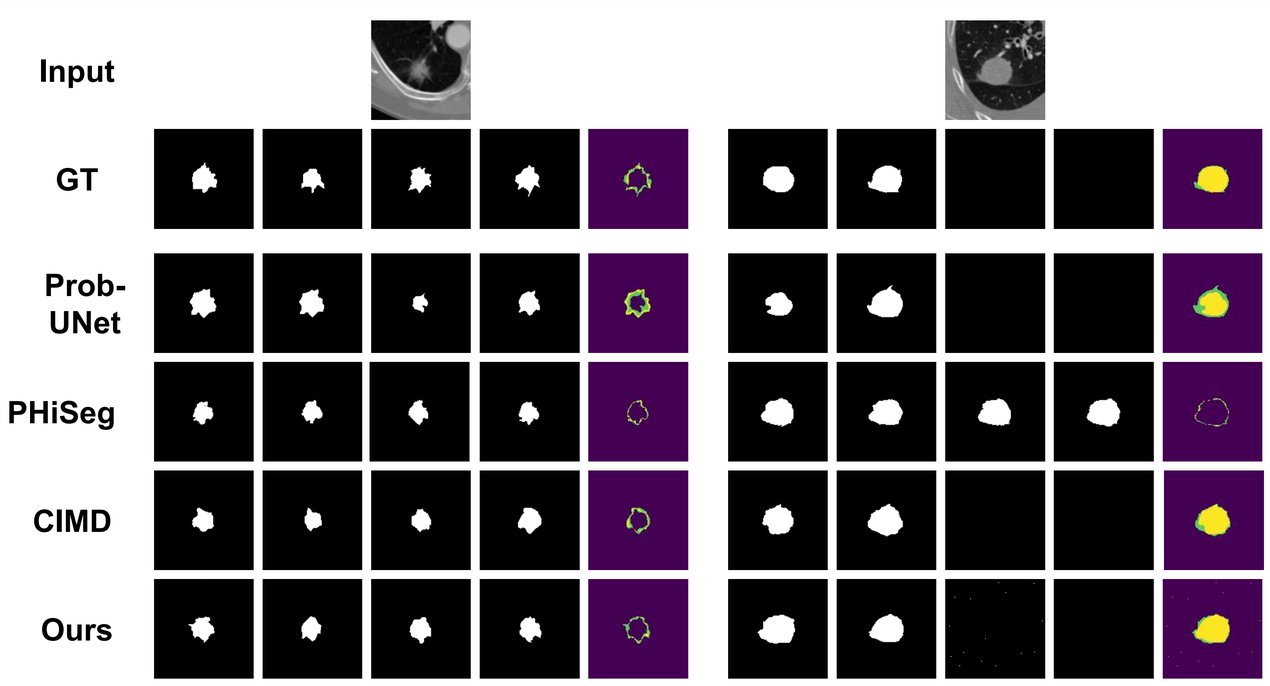


Figure 1. Qualitative comparison on LIDC-IDRI: input images, followed by segmentation masks and uncertainty maps for GT, baselines, and our method.

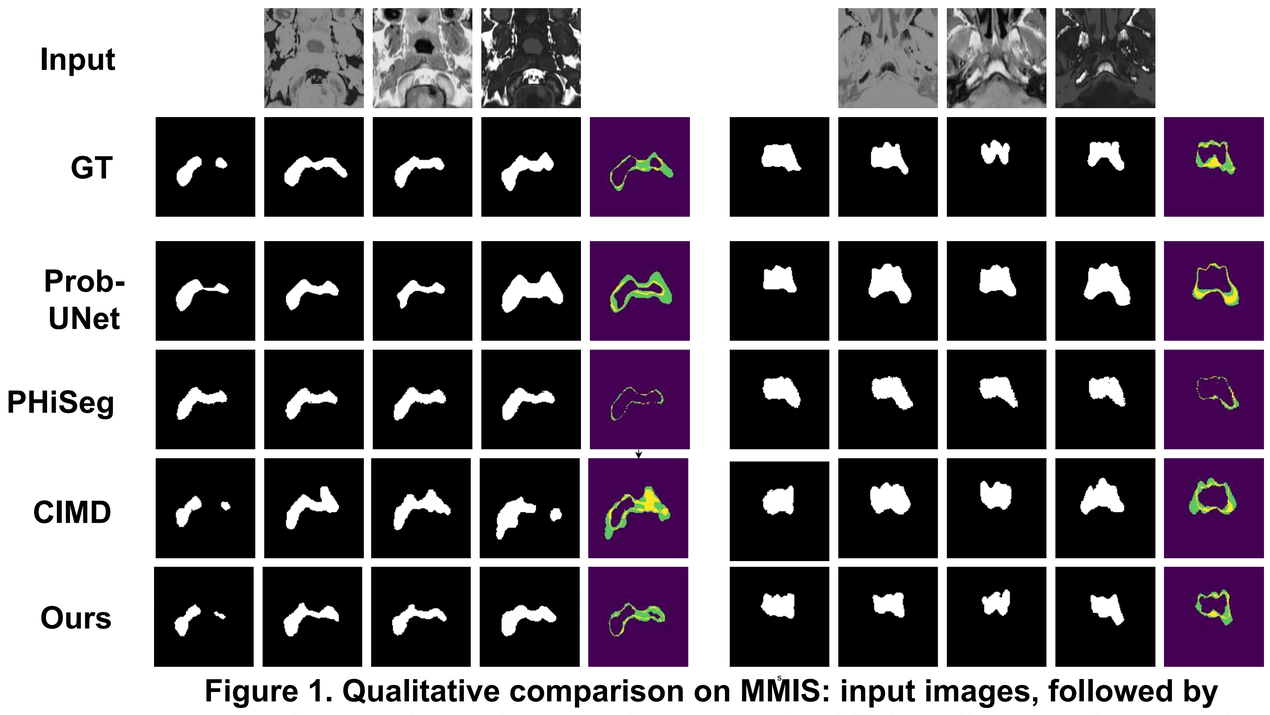


Figure 1. Qualitative comparison on MMIS: input images, followed by segmentation masks and uncertainty maps for GT, baselines, and our method.

Dataset	Method	N_{sample}	GED ↓	S_{NCC} ↑	$D_{\max}$ ↑	Dice ↑
LIDC-IDRI	Prob-UNet	5	0.353	0.750	0.787	0.505
		10	0.300	0.817	0.859	0.520
		15	**0.290**	0.831	0.870	0.525
	PHiSeg	5	0.533	0.310	0.565	0.579
		10	0.518	0.342	0.578	0.581
		15	0.513	0.358	0.586	0.577
	CIMD	5	0.395	0.791	0.778	0.556
		10	0.339	0.828	0.835	0.584
		15	0.297	0.830	0.870	0.556
	(Ours)	5	0.409	0.816	0.739	0.537
		10	0.323	0.832	0.848	0.578
		15	0.292	**0.835**	**0.876**	**0.595**
MMIS	Prob-UNet	5	0.280	0.501	0.813	0.729
		10	0.240	0.574	0.845	0.738
		15	**0.227**	0.607	0.856	0.740
	PHiSeg	5	0.328	0.217	0.749	0.770
		10	0.321	0.243	0.755	0.776
		15	0.319	0.254	0.757	0.780
	CIMD	5	0.301	0.568	0.797	0.765
		10	0.256	0.695	0.834	0.773
		15	0.235	0.742	0.849	0.780
	(Ours)	5	0.294	0.816	0.813	0.770
		10	0.244	0.832	0.842	0.778
		15	0.231	**0.835**	**0.856**	**0.785**

Table 1. Quantitative evaluation of our model on two datasets versus three baselines, with increasing N_{sample}.

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